Pre−Pandemic Data Only

antiGen
ESM 2 (35M)
BLOSUM62
EVEscape
ESM 2 (650M)
Codon table

antiGen Finetunes

antiGen
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) partial FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as EVE full FT

Pandemic Data in Training Window

antiGen
PSSM Conservation
MSA Trans. (BA.2 eval)
Tranception
Substitution matrix
PSSM
MSA Transfomer
EVE
EVmutation
Where × which

Deep Mutational Scans

antiGen
BA.1 DMS (expression)
BA.2 DMS (expression)
BA.1 DMS (bind)
BA.2 DMS (bind)

all mutations, 1+ cases

de novo, 1+ cases

all mutations, 100+ cases

de novo, 100+ cases

A B C D E F G H I J K L M N O P

P(mut. in top decile obs. by t)

Date (t, years into pandemic)